SUPERSTORE SALES ANALYSIS USING SQL

Celebal Summer Internship 2026

Week 3 Assignment

Sales Data Analysis using Subqueries, CTEs and Window Functions

Dataset: Sample Superstore Dataset

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SETUP DATA

Exploring Database

USE SuperstoreDB

Check total records in the dataset

Query:

SELECT COUNT(*) FROM superstore_raw;

Output:

	(No column name)
1	9994

View sample records from the raw dataset

Query:

SELECT TOP 10 * FROM superstore_raw;

Output:

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.95999
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro...	731.94000
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Univ...	14.619999
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.57751
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368000
6	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessor...	48.860000
7	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.2800002
8	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.15197

CREATING Customers TABLE

Query:

```
CREATE TABLE customers

(

  customer_id VARCHAR(50),

  customer_name VARCHAR(100),

  segment VARCHAR(50),
```

```
  country VARCHAR(50),
  city VARCHAR(50),
  state VARCHAR(50),
  postal_code INT,
  region VARCHAR(50)
);
```

INSERTING INTO Customers TABLE

Query:

```
INSERT INTO customers
SELECT DISTINCT
  Customer_ID,
  Customer_Name,
  Segment,
  Country,
  City,
  State,
  Postal_Code,
  Region
FROM superstore_raw;
```

Displaying customer data :

Query:

```
SELECT * from customers;
```

Output:

	customer_id	customer_name	segment	country	city	state	postal_code	region
1	AA-10315	Alex Avila	Consumer	United States	Minneapolis	Minnesota	55407	Central
2	AA-10315	Alex Avila	Consumer	United States	New York City	New York	10011	East
3	AA-10315	Alex Avila	Consumer	United States	Round Rock	Texas	78664	Central
4	AA-10315	Alex Avila	Consumer	United States	San Francisco	California	94109	West
5	AA-10315	Alex Avila	Consumer	United States	San Francisco	California	94122	West
6	AA-10375	Allen Armold	Consumer	United States	Atlanta	Georgia	30318	South
7	AA-10375	Allen Armold	Consumer	United States	Lebanon	Tennessee	37087	South
8	AA-10375	Allen Armold	Consumer	United States	Los Angeles	California	90008	West
9	AA-10375	Allen Armold	Consumer	United States	Mesa	Arizona	85204	West

CREATING Products TABLE

<u>Query:</u>

```
CREATE TABLE products
(
   product_id VARCHAR(50),
   category VARCHAR(50),
   sub_category VARCHAR(50),
   product_name VARCHAR(255)
);
```

INSERTING INTO Products TABLE

<u>Query:</u>

```
INSERT INTO products
SELECT DISTINCT
   Product_ID,
   Category,
   Sub_Category,
   Product_Name
FROM superstore_raw;
```

Display Products data

Query:

SELECT * FROM products

Output:

	product_id	category	sub_category	product_name
1	FUR-BO-10000112	Furniture	Bookcases	Bush Birmingham Collection Bookcase, Dark Cherry
2	FUR-BO-10000330	Furniture	Bookcases	Sauder Camden County Barrister Bookcase, Planked ...
3	FUR-BO-10000362	Furniture	Bookcases	Sauder Inglewood Library Bookcases
4	FUR-BO-10000468	Furniture	Bookcases	O'Sullivan 2-Shelf Heavy-Duty Bookcases
5	FUR-BO-10000711	Furniture	Bookcases	Hon Metal Bookcases, Gray
6	FUR-BO-10000780	Furniture	Bookcases	O'Sullivan Plantations 2-Door Library in Landvery Oak
7	FUR-BO-10001337	Furniture	Bookcases	O'Sullivan Living Dimensions 2-Shelf Bookcases
8	FUR-BO-10001519	Furniture	Bookcases	O'Sullivan 3-Shelf Heavy-Duty Bookcases
9	FUR-BO-10001567	Furniture	Bookcases	Bush Westfield Collection Bookcases, Dark Cherry Fi

CREATING Orders TABLE

Query :

```
CREATE TABLE orders

(

   row_id INT PRIMARY KEY,

   order_id VARCHAR(50),

   order_date DATE,

   ship_date DATE,

   ship_mode VARCHAR(50),

   customer_id VARCHAR(50),

   product_id VARCHAR(50),

   sales DECIMAL(10,2),

   quantity INT,

   discount DECIMAL(5,2),

   profit DECIMAL(10,2)

);
```

INSERTING INTO Orders TABLE

<u>Query :</u>

```
INSER INTO orders
SELECT DISTINCT
  Row_ID,
  Order_ID,
  Order_Date,
  Ship_Date,
  Ship_Mode,
  Customer_ID,
  Product_ID,
  Sales,
  Quantity,
  Discount,
  Profit
FROM superstore_raw;
```

Displaying orders data

```
SELECT * FROM orders
```

<u>Output:</u>

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0.00	41.91
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0.00	219.58
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	OFF-LA-10000240	14.62	2	0.00	6.87
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	FUR-TA-10000577	957.58	5	0.45	-383.03
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	OFF-ST-10000760	22.37	2	0.20	2.52
6	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	FUR-FU-10001487	48.86	7	0.00	14.17
7	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	OFF-AR-10002833	7.28	4	0.00	1.97
8	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	TEC-PH-10002275	907.15	6	0.20	90.72
9	9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	OFF-BI-10003910	18.50	3	0.20	5.78

Verify data inserted into all tables

<u>Query :</u>

SELECT TOP 5 * FROM customers;

SELECT TOP 5 * FROM products;

SELECT TOP 5 * FROM orders;

Output:

	customer_id	customer_name	segment	country	city	state	postal_code	region
1	AA-10315	Alex Avila	Consumer	United States	Minneapolis	Minnesota	55407	Central
2	AA-10315	Alex Avila	Consumer	United States	New York City	New York	10011	East
3	AA-10315	Alex Avila	Consumer	United States	Round Rock	Texas	78664	Central
4	AA-10315	Alex Avila	Consumer	United States	San Francisco	California	94109	West
5	AA-10315	Alex Avila	Consumer	United States	San Francisco	California	94122	West

	product_id	category	sub_category	product_name
1	FUR-BO-10000112	Furniture	Bookcases	Bush Birmingham Collection Bookcase, Dark Cherry
2	FUR-BO-10000330	Furniture	Bookcases	Sauder Camden County Barrister Bookcase, Planked ...
3	FUR-BO-10000362	Furniture	Bookcases	Sauder Inglewood Library Bookcases
4	FUR-BO-10000468	Furniture	Bookcases	O'Sullivan 2-Shelf Heavy-Duty Bookcases
5	FUR-BO-10000711	Furniture	Bookcases	Hon Metal Bookcases, Gray

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0.00	41.91
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0.00	219.58
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	OFF-LA-10000240	14.62	2	0.00	6.87
4	4	US-2015-1089...	2015-10-11	2015-10-18	Standard Cl...	SO-20335	FUR-TA-10000577	957.58	5	0.45	-383....
5	5	US-2015-1089...	2015-10-11	2015-10-18	Standard Cl...	SO-20335	OFF-ST-10000760	22.37	2	0.20	2.52

Section 1: Subquery

Question 1. Find all orders where sales are greater than the average sales

Query :

Select

order_id,

customer_id,

sales

From orders Where sales > (Select AVG(Sales) from orders)

ORDER BY sales DESC

Output:

	order_id	customer_id	sales
1	CA-2014-145317	SM-20320	22638.48
2	CA-2016-118689	TC-20980	17499.95
3	CA-2017-140151	RB-19360	13999.96
4	CA-2017-127180	TA-21385	11199.97
5	CA-2017-166709	HL-15040	10499.97
6	CA-2016-117121	AB-10105	9892.74
7	CA-2014-116904	SC-20095	9449.95
8	US-2016-107440	BS-11365	9099.93
9	CA-2016-158841	SE-20110	8749.95
10	CA-2016-143714	CC-12370	8399.98
11	CA-2014-143917	KL-16645	8187.65

<u>Question 2.</u> Find the highest sales order for each customer

<u>Query :</u>

Select

order_id,

customer_id,

sales

FROM orders o where sales =

(Select MAX(Sales) from orders where customer_id = o.customer_id)

<u>Output:</u>

	order_id	customer_id	sales
1	CA-2016-152156	CG-12520	731.94
2	US-2015-108966	SO-20335	957.58
3	CA-2014-115812	BH-11710	1706.18
4	CA-2015-106320	EB-13870	1044.63
5	US-2015-150630	TB-21520	3083.43
6	CA-2016-117590	GH-14485	1097.54
7	CA-2015-117415	SN-20710	532.40
8	CA-2016-105816	JM-15265	1029.95
9	CA-2016-111682	TB-21055	319.41
10	CA-2014-106376	BS-11590	1113.02
11	CA-2015-129476	PA-19060	339.96

Section 2 : CTE

<u>Question 3</u>. Calculate total sales for each customer

Query :

```
WITH customer_sales AS
(
  SELECT
    customer_id,
    SUM(sales) AS total_sales
  FROM orders
  GROUP BY customer_id
)
SELECT *
FROM customer_sales
ORDER BY total_sales DESC;
```

Output:

	customer_id	total_sales
1	SM-20320	25043.07
2	TC-20980	19052.22
3	RB-19360	15117.35
4	TA-21385	14595.62
5	AB-10105	14473.57
6	KL-16645	14175.23
7	SC-20095	14142.34
8	HL-15040	12873.30
9	SE-20110	12209.44
10	CC-12370	12129.08
11	TS-21370	11891.75
12	GT-14710	11820.12
13	BM-11140	11789.64
14	SV-20365	11470.94
15	CJ-12010	11164.97
16	CL-12565	10880.56
17	ME-17320	10663.73

Question 4. Find customers whose total sales are above average

Query :

```sql
WITH customer_sales AS
(
  SELECT
    customer_id,
    SUM(sales) AS total_sales
  FROM orders
  GROUP BY customer_id
)SELECT
  customer_id,
  total_sales,
  (SELECT AVG(total_sales) FROM customer_sales) AS average_sales
FROM customer_sales
WHERE total_sales >
(
  SELECT AVG(total_sales)
  FROM customer_sales
);
```

<u>Output:</u>

	customer_id	total_sales	average_sales
1	LF-17185	3930.52	2896.848373
2	KH-16330	3312.87	2896.848373
3	SC-20305	3979.06	2896.848373
4	SV-20785	4105.32	2896.848373
5	JL-15505	3173.87	2896.848373
6	RB-19360	15117.35	2896.848373
7	FP-14320	4046.75	2896.848373
8	HR-14830	5248.80	2896.848373
9	JM-15250	3159.11	2896.848373
10	SR-20740	4345.88	2896.848373
11	DG-13300	3195.82	2896.848373
12	MC-17845	3805.71	2896.848373

Section 3 : Window Function

1. RANK() :

<u>Question 5.</u> Rank all customers based on total sales

<u>Query:</u>

```
SELECT
customer_id,
SUM(sales) AS total_sales,
RANK() OVER (ORDER BY SUM(sales) DESC) AS customer_rank
FROM orders
GROUP BY customer_id;
```

<u>Ouput:</u>

	customer_id	total_sales	customer_rank
1	SM-20320	25043.07	1
2	TC-20980	19052.22	2
3	RB-19360	15117.35	3
4	TA-21385	14595.62	4
5	AB-10105	14473.57	5
6	KL-16645	14175.23	6
7	SC-20095	14142.34	7
8	HL-15040	12873.30	8
9	SE-20110	12209.44	9
10	CC-12370	12129.08	10
11	TS-21370	11891.75	11
12	GT-14710	11820.12	12
13	BM-11140	11789.64	13
14	SV-20365	11470.94	14
15	CJ-12010	11164.97	15
16	CL-12565	10880.56	16

2 . ROW_NUMBER() :

<u>Question 6.</u> Assign row numbers to each order within a customer.

<u>Query:</u>

```
SELECT
```

customer_id,

order_id,

sales,

ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY sales DESC) AS row_num

FROM orders;

Output:

	customer_id	order_id	sales	row_num
1	AA-10315	CA-2016-103982	3930.07	1
2	AA-10315	CA-2014-128055	673.57	2
3	AA-10315	CA-2016-103982	431.98	3
4	AA-10315	CA-2017-147039	362.94	4
5	AA-10315	CA-2014-128055	52.98	5
6	AA-10315	CA-2016-103982	41.72	6
7	AA-10315	CA-2015-121391	26.96	7
8	AA-10315	CA-2014-138100	14.94	8
9	AA-10315	CA-2014-138100	14.56	9
10	AA-10315	CA-2017-147039	11.54	10
11	AA-10315	CA-2016-103982	2.30	11
12	AA-10375	CA-2016-131065	499.98	1
13	AA-10375	CA-2015-140921	149.97	2

Question 7. Display top 3 customers based on total sales.

Query :

SELECT * FROM

(SELECT

customer_id,

SUM(sales) AS total_sales,

RANK() OVER(ORDER BY SUM(sales) DESC) AS customer_rank

FROM orders GROUP BY customer_id

) t WHERE customer_rank <= 3;

Output:

	customer_id	total_sales	customer_rank
1	SM-20320	25043.07	1
2	TC-20980	19052.22	2
3	RB-19360	15117.35	3

Section 5 : JOIN + CTE + Window Function

Question : Customer sales summary with ranking

Query :

```
with customer_sales AS
(
  SELECT
    customer_id,
    SUM(sales) AS total_sales
  FROM orders
  GROUP BY customer_id
)
SELECT
  c.customer_name,
  cs.total_sales,
  RANK() OVER(ORDER BY cs.total_sales DESC) AS customer_rank
FROM customer_sales cs
JOIN customers c ON cs.customer_id = c.customer_id;
```

Output:

	customer_name	total_sales	customer_rank
1	Sean Miller	25043.07	1
2	Sean Miller	25043.07	1
3	Sean Miller	25043.07	1
4	Sean Miller	25043.07	1
5	Sean Miller	25043.07	1
6	Tamara Chand	19052.22	6
7	Tamara Chand	19052.22	6
8	Tamara Chand	19052.22	6
9	Tamara Chand	19052.22	6
10	Tamara Chand	19052.22	6
11	Raymond Buch	15117.35	11
12	Raymond Buch	15117.35	11
13	Raymond Buch	15117.35	11

MINI Project

Section 6 : BUSINESS Queries (Customer Sales Insights)

1. Who are the top 5 customers?

<u>Query :</u>

```
WITH customer_sales AS
(
  SELECT
    customer_id,
    SUM(sales) AS total_sales
  FROM orders GROUP BY customer_id
)
SELECT TOP 5
  customer_id,
  total_sales
FROM customer_sales ORDER BY total_sales DESC;
```

<u>Output:</u>

	customer_id	total_sales
1	SM-20320	25043.07
2	TC-20980	19052.22
3	RB-19360	15117.35
4	TA-21385	14595.62
5	AB-10105	14473.57

2. Who are the bottom 5 customers?

Query :

```
WITH customer_sales AS
(
  SELECT
    customer_id,
    SUM(sales) AS total_sales
  FROM orders GROUP BY customer_id
)
SELECT TOP 5
  customer_id,
  total_sales
FROM customer_sales ORDER BY total_sales ASC;
```

Output:

	customer_id	total_sales
1	TS-21085	4.84
2	LD-16855	5.30
3	CJ-11875	16.52
4	MG-18205	16.74
5	RS-19870	22.33

3. Which customers made only one order?

Query :

```
SELECT
```

customer_id,

COUNT(DISTINCT order_id) AS total_orders

FROM orders

GROUP BY customer_id

HAVING COUNT(DISTINCT order_id) = 1;

Output:

	customer_id	total_orders
1	AO-10810	1
2	AR-10570	1
3	CJ-11875	1
4	JC-15385	1
5	JR-15700	1
6	LD-16855	1
7	MG-18205	1
8	PH-18790	1
9	RE-19405	1
10	RM-19750	1
11	SM-20905	1
12	TC-21145	1

4. Which customers have above-average sales?

Query :

WITH customer_sales AS

(

SELECT

customer_id,

SUM(sales) AS total_sales

FROM orders

GROUP BY customer_id

)

SELECT

customer_id,

```
    total_sales,
    (SELECT AVG(total_sales) FROM customer_sales) AS average_sales
FROM customer_sales
WHERE total_sales >
(
    SELECT AVG(total_sales)
    FROM customer_sales
);
```

Output:

	customer_id	total_sales	average_sales
1	LF-17185	3930.52	2896.848373
2	KH-16330	3312.87	2896.848373
3	SC-20305	3979.06	2896.848373
4	SV-20785	4105.32	2896.848373
5	JL-15505	3173.87	2896.848373
6	RB-19360	15117.35	2896.848373
7	FP-14320	4046.75	2896.848373
8	HR-14830	5248.80	2896.848373
9	JM-15250	3159.11	2896.848373
10	SR-20740	4345.88	2896.848373

Question 5 : What is the highest order value per customer?

Query:

```
SELECT
    order_id,
    customer_id,
    sales AS highest_order_value
FROM orders o
WHERE sales =
(
    SELECT MAX(sales)
    FROM orders
```

```
    WHERE customer_id = o.customer_id
);
```

Output:

	order_id	customer_id	highest_order_value
1	CA-2016-152156	CG-12520	731.94
2	US-2015-108966	SO-20335	957.58
3	CA-2014-115812	BH-11710	1706.18
4	CA-2015-106320	EB-13870	1044.63
5	US-2015-150630	TB-21520	3083.43
6	CA-2016-117590	GH-14485	1097.54
7	CA-2015-117415	SN-20710	532.40
8	CA-2016-105816	JM-15265	1029.95
9	CA-2016-111682	TB-21055	319.41
10	CA-2014-106376	BS-11590	1113.02
11	CA-2015-129476	PA-19060	339.96
12	CA-2015-110744	HA-14920	671.93
13	CA-2016-114489	JE-16165	1951.84